Haftom Alemayehu

Senior Full-Stack Software Engineer | AI-Assisted Engineering | Platform Enablement | Cloud-Native Architecture | Onshore/Offshore Technical Leadership

PROFESSIONAL SUMMARY

Hands-on Senior Full Stack Software Engineer and Solution Architect with 12+ years of experience architecting, designing, developing, and deploying highly available, secure, and scalable cloud-native enterprise applications. Experienced in modern Generative AI engineering, including AI agent development, Retrieval-Augmented Generation (RAG) solutions, and enterprise adoption of LLM-powered platforms such as ChatGPT, Microsoft Copilot, GitHub Copilot, Claude Code, and Cursor. Expertise includes Java, Spring Boot, TypeScript, React, Angular, GraphQL, REST APIs, AWS, Kubernetes, OpenShift, Terraform, CI/CD automation, microservices architecture, API management, and AI-enabled software engineering workflows. Experienced in designing AI solutions, defining proof-of-concept success criteria, establishing AI performance KPIs, implementing production monitoring strategies, and integrating GenAI capabilities into enterprise applications.

CORE COMPETENCIES

Generative AI & AI Engineering: ChatGPT, Microsoft Copilot, GitHub Copilot, Claude Code, Cursor, Generative AI, LLMs, AI Agents, Agentic AI Workflows, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search, Embeddings, AI Evaluation Metrics, AI Observability

AI Platforms & Integration: LLM APIs, Enterprise AI Integration, AI Workflow Automation, Knowledge Retrieval Systems, AI-powered Developer Tools, Cloud AI Services

AI-Assisted Engineering: Claude Code, Cursor, GitHub Copilot, Devin AI (code generation, automated unit tests, code review, documentation) — used for day-to-day implementation, architectural reasoning, repository-wide refactoring, and autonomous feature implementation.

Cloud & Containers: AWS (EKS, ECS, ECR, API Gateway, S3, RDS, DynamoDB, SQS, IAM, KMS, CloudWatch), Docker, Kubernetes, OpenShift, GCP, Azure

APIs & Messaging: REST Web Services, GraphQL (Apollo Federation, schema design, resolvers, subscriptions), Apigee, OpenAPI/Swagger, Kafka, SOAP, JSON & XML technologies / parsers

Design Patterns & Security: MVC, Singleton, Factory, Facade, DAO, DTO, Session; domain-driven & 12-Factor design; OAuth 2.0, OIDC, JWT, Vault, OWASP, secure SDLC

Core Skills: Java 8 / 11 / 17 / 21, Python, TypeScript, JavaScript (ES6+), Node.js, SQL, REST APIs, AWS Lambda, API Gateway, Angular, Terraform, Docker, Kubernetes, PostgreSQL

Spring Framework: Spring Core, Spring MVC, Spring Boot, Spring Security, Spring Data (JPA), Dependency Injection, DAO, JDBC, JTA, transaction management; ORM: Hibernate

Databases & SQL: PostgreSQL, MS SQL Server, Oracle, MySQL/MariaDB; advanced SQL (complex JOINs, aggregation, window functions), query optimization & performance tuning, indexing, execution-plan analysis; NoSQL: MongoDB, DynamoDB, Cassandra

Frontend: Angular 9+, TypeScript, React.js, Redux, JavaScript (ES6+), HTML5, jQuery, CSS3/SASS, responsive UI

DevOps / CI-CD: Jenkins, GitHub Actions, GitLab CI/CD, Harness, AWS CodePipeline, Azure DevOps, JIRA, Terraform, secure SDLC (SAST/DAST), Vault, Splunk, Prometheus/Grafana

Testing: JUnit, Mockito, Cucumber (BDD), TDD, integration testing

Microservices: Spring Boot Microservices, REST APIs, GraphQL, Event-Driven Architecture, Kafka Streaming, API Security, OAuth 2.0, JWT, mTLS, Idempotency, Retries, Backoff, Circuit Breakers

CONTACT

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KEY STRENGTHS

- Cloud-native microservices architecture
- Enterprise application modernization
- API design & integration (REST/GraphQL)
- DevSecOps & CI/CD automation
- Performance tuning & optimization
- Onshore/offshore technical leadership
- AI-assisted engineering workflows

EDUCATION

M.S., Computer Software Engineering
Maharishi University of Management
Fairfield, IA · 2015–2018

M.S., Strategic IT Management
Stockholm University
Stockholm, Sweden · 2011–2013

B.S., Information Technology Engineering
Mekelle Institute of Technology
Mekelle, Ethiopia · 2005–2009

CERTIFICATIONS

Kubernetes — Package Management with Helm
Generative AI (foundations)
AWS Solutions Architect – Professional (in progress)

LANGUAGES

English (Fluent)
Swedish (Professional Working)
Amharic (Native)
Tigrinya (Native)

Data Engineering & Integration: ETL/ELT, ESB, EAI, Batch Processing, Near Real-Time Streaming, CDC Concepts, Data Integration; modeling: BPMN, UML, ER

PROFESSIONAL EXPERIENCE

Integration Solution Architect & Team Lead **10/2023 – Present**

IHG Hotels & Resorts · Charlotte, United States

- Provided technical leadership for a cross-functional team of 15 across distributed onshore/offshore engineering teams in the United States, India, and Europe — establishing development standards, leading architecture reviews and pair-programming sessions, mentoring engineers and accelerating onboarding, owning sprint planning, backlog refinement, technical estimation, and release coordination, and managing technical risks and cross-team dependencies while driving adoption of cloud-native, AI-assisted engineering practices across multiple product teams.
- Partnered with executive leadership, product owners, architects, security, and infrastructure teams to define technical strategy, presenting architecture decisions, technical roadmaps, and modernization strategies to senior leadership and business partners.
- Architected and developed Java 17/Spring Boot backend microservices, exposing secure RESTful and GraphQL APIs for distributed, cloud-native, high-availability platforms, deployed and managed on Red Hat OpenShift using Kubernetes, Docker, Helm, and CI/CD pipelines to improve scalability, resilience, and deployment automation.
- Designed asynchronous, event-driven workflows using Kafka, SQS, and messaging patterns for scalable distributed transaction processing, and developed serverless Java functions with Spring Cloud Function that run the same Spring Boot logic as standalone microservices and cloud functions, improving reuse and testability.
- Architected and implemented platform services standardizing authentication, configuration management, and deployment patterns on Kubernetes, with resiliency patterns—retries, exponential backoff, circuit breakers, timeout controls, pagination, idempotency, and fault isolation—plus distributed tracing and health monitoring to improve observability and simplify production debugging.
- Performed root cause analysis using Splunk to troubleshoot production incidents across microservices, APIs, Kubernetes clusters, messaging systems, databases, and cloud infrastructure, and provided documentation and knowledge transfer enabling engineering teams to use Splunk for monitoring, troubleshooting, and operational analytics.
- Engineered and optimized advanced SQL queries (complex multi-table JOINs, aggregation, window functions) on PostgreSQL and RDS, analyzing execution plans and implementing indexing strategies to maintain low latency under peak load, and implemented transactional data-access layers using Spring Data (JPA), Hibernate, JDBC, and JTA to enforce data integrity and consistency across services.
- Constructed responsive frontend components using Angular/TypeScript and React, seamlessly integrating with backend REST and GraphQL services, while optimizing rendering, lazy loading, and state management.
- Spearheaded the architectural design of a Backend-for-Frontend (BFF) layer utilizing GraphQL and Apollo Federation, consolidating multiple internal and third-party services behind a centralized, secured gateway, and established enterprise API standards and governance (authentication/authorization, versioning, logging & monitoring, error handling, and API lifecycle management).
- Designed reusable enterprise API services and shared platform capabilities that eliminated duplicated business logic across applications and accelerated future product development, adopting emerging interoperability standards including Model Context Protocol (MCP) for AI-tool and service integration.
- Designed target-state architecture and phased migration roadmaps for legacy enterprise applications transitioning to cloud-native microservices.
- Integrated enterprise commercial off-the-shelf (COTS) and SaaS platforms, enterprise APIs, identity providers, payment services, and third-party vendor solutions into cloud-native systems.

- Led adoption of AI-assisted engineering practices across multiple product teams using Claude Code, Cursor, GitHub Copilot, and Devin AI, developing reusable prompts, engineering standards, and workflow templates that accelerated code generation, automated testing, documentation, and code reviews while improving delivery velocity and developer productivity.
- Established enterprise AI engineering standards (prompt engineering, code-review workflows, testing, documentation) and defined AI adoption playbooks, leading phased enterprise rollout with reusable templates and internal workshops to enable teams beyond individual mentoring.
- Developed automation solutions using Python, Bash, and TypeScript to provision cloud infrastructure, orchestrate Kubernetes deployments, automate operational tasks, integrate with CI/CD pipelines, and improve developer productivity across AWS and Kubernetes platforms.
- Deployed scalable, secure cloud-native solutions on AWS (EKS, ECS, API Gateway, S3, IAM, DynamoDB, SQS, KMS) with a multi-tier VPC architecture—private-subnet EKS microservices behind public load balancers, isolated RDS, and security groups, route tables, NAT/Internet Gateways, and VPC endpoints enforcing least-privilege access—using reusable Terraform modules and GitOps-driven pipelines for repeatable, secure infrastructure deployment.
- Managed end-to-end delivery lifecycle, encompassing analysis, design, development, automated testing (JUnit, Mockito, Cucumber), and deployment through Jenkins/GitHub Actions, introducing automated quality gates and secure SDLC practices that reduced deployment failures and improved release confidence.
- Established reusable backend development patterns that reduced implementation time for new microservices by approximately 30%.
- Designed data integration workflows spanning batch processing, near-real-time streaming, and database synchronization using Kafka-based event pipelines and enterprise data services, including log-based change data capture (CDC) to synchronize transactional data across distributed systems.
- Implemented API security controls including OAuth 2.0/JWT authentication, mTLS service-to-service communication, rate limiting, throttling, and idempotent transaction processing.

Senior Software Engineer (Full Stack, Backend-Heavy) **07/2020 – 10/2023**

Wells Fargo · Charlotte, United States

- Delivered robust Java/Spring Boot backend services and transaction-processing flows for high-volume, regulated financial platforms, prioritizing data integrity, security, and auditability, and collaborating with compliance, security, and risk management teams to meet regulatory requirements.
- Supported production releases, incident response, architecture reviews, and critical deployment activities for enterprise applications in high-availability environments.
- Engineered and maintained microservices on OpenShift, leveraging Kubernetes orchestration, rolling deployments, health probes, and autoscaling to improve application reliability.
- Designed asynchronous, event-driven workflows using Kafka, SQS, and Spring Cloud Function components—consuming Kafka messages, performing validation and transformation, and invoking downstream REST APIs—to support scalable distributed transaction processing across banking platforms.
- Automated OpenShift provisioning and deployments via Jenkins and GitHub Actions with reusable Terraform modules (VPCs, IAM roles, Kubernetes clusters, subnets, security groups, NAT/Internet Gateways, VPC endpoints) and policy-driven validation, standardizing Infrastructure as Code across environments with least-privilege networking and reducing setup time from days to under an hour.
- Executed performance tuning for an enterprise order/transaction-processing system; optimized lengthy reporting queries over large transactional datasets by analyzing execution plans, redesigning queries, implementing targeted indexes, applying aggregation strategies, and introducing caching, reducing report generation time from minutes to seconds and significantly alleviating database load during peak business hours.

- Authored sophisticated SQL queries (complex JOINs, subqueries, aggregation) for MS SQL Server, Oracle, and PostgreSQL, overseeing schema design, indexing, and query optimization.
- Built secure, resilient RESTful and GraphQL APIs using Spring MVC, Spring Security, Spring Data (JPA), and Hibernate—enforcing OAuth 2.0/OIDC/JWT authentication, mTLS, rate limiting, and throttling, with retry mechanisms, exponential backoff, circuit breakers, pagination, timeout controls, and idempotent transaction processing—within compliance-driven environments.
- Pioneered the migration from REST-based integrations to GraphQL, achieving approximately a 70% reduction in API calls and establishing a unified contract for Angular/React and mobile development teams.
- Explored early LLM-assisted development in the role's later period (2022–2023) — adopting GitHub Copilot and generative AI tooling and designing AI-assisted development and early agentic patterns for feature implementation, code analysis, troubleshooting, and documentation, and applying RAG concepts to improve access to technical documentation and enterprise information.
- Created responsive frontend features in Angular/TypeScript and React, integrating them with backend services and collaborating closely with UX/UI and Product stakeholders.
- Designed, managed, and secured end-to-end API solutions with Apigee, encompassing proxy creation, access control, monitoring, and publishing, utilizing OAuth 2.0, OIDC, and JWT for service security.
- Containerized services using Docker and orchestrated them on Kubernetes, implementing CI/CD pipelines (Jenkins, GitHub Actions) and secure SDLC practices (SAST/DAST, Vault) within compliance-driven environments.
- Performed root cause analysis with Splunk across production incidents (microservices, APIs, Kubernetes, messaging, databases, cloud infrastructure), reducing resolution times, and provided documentation and knowledge transfer enabling teams to leverage Splunk for monitoring and operational analytics.
- Built data integration pipelines spanning batch processing, near-real-time streaming, and database synchronization, including log-based change data capture (CDC) to keep transactional data consistent across distributed systems.
- Developed Python utilities for scripting, automation, API integrations, and data processing.

Lead Software Engineer **01/2020 – 06/2020**

Duke Energy Corporation · Charlotte, United States

- Architected, built, and tested Java/Spring Boot microservices within an Agile/Scrum framework, modernizing legacy monolithic systems into domain-driven, bounded-context services.
- Engineered cloud-native applications utilizing Java 8+, Spring Boot, and Spring Cloud, supporting continuous deployment pipelines and securing/monitoring APIs via Apigee.
- Collaborated with DevOps teams to containerize applications and deploy them to OpenShift, streamlining release management, environment consistency, and operational efficiency.
- Automated provisioning of Kubernetes clusters, networking, and cloud services using Terraform, creating reusable Infrastructure as Code templates that improved deployment consistency and reduced configuration drift across environments.
- Developed Spring Cloud Function-based serverless Java components and configured AWS VPC networking (subnets, security groups, NAT gateways, VPC endpoints) to support secure, cloud-native microservice deployments.
- Supported production OpenShift clusters by troubleshooting deployments, monitoring application health, and optimizing resource utilization.

Senior Software Engineer **06/2019 – 01/2020**

Bank of America · Charlotte, United States

- Migrated legacy applications to OpenShift, enabling cloud-native architecture and zero-downtime deployments.

- Built event-driven microservices leveraging Kafka and delivered GraphQL APIs, enforcing authentication and authorization through OIDC and OAuth 2.0 protocols.
- Led triage and root-cause analysis of production incidents during on-call support, driving fixes that restored service and reduced recurrence.
- Built, configured, and deployed customer-facing financial applications utilizing Spring Boot and RESTful services within a microservices-oriented Agile team.

Senior Software Engineer **10/2018 – 05/2019**

Tata Consultancy Services · Charlotte, United States

- Practiced Test-Driven Development (TDD) with a strong focus on modular design and comprehensive unit/integration testing (JUnit), developing reusable React.js components from provided designs and wireframes.
- Integrated with backend SOAP and REST web services, designing, securing, and monitoring APIs through the Apigee platform.
- Architected microservice-oriented solutions using Spring Boot, Spring Cloud, Docker, and Apache Camel, building resilient and highly available cloud-native services.
- Created user-facing features in React.js, applied security measures with OAuth 2.0, JWT, and Apigee, and successfully decomposed legacy applications into cohesive, bounded-context services.

Senior Software Developer **08/2016 – 10/2018**

University of Iowa Hospitals & Clinics · Iowa City, United States

- Supported enterprise application engineering for a large academic healthcare system, designing and developing scalable, secure, distributed enterprise applications using Java 8, Spring Boot, and Hibernate.
- Modernized legacy application components, reducing the codebase by 75%, and introduced 12-Factor application development principles to improve maintainability and scalability.
- Developed reusable React.js components that improved user-experience scores by 50%, and implemented OAuth 2.0 and JWT for secure application access.

EARLIER EXPERIENCE

- Swedbank AB — Senior Software Developer · Stockholm · 2014–2015
- SWED-Eth Chamber of Commerce — Software Developer · Stockholm · 2014
- Maria Nila AB — E-Commerce Developer · Stockholm · 2013–2014 — Grew online sales by 80% through a custom eCommerce platform.
- INSA, Governmental R&D — Software Developer · Ethiopia · 2009–2011